Assignment – II

1) Write a program that takes a score (0-100) and prints:
   "A" if score >= 90
   "B" if score >= 80
   "C" if score >= 70
   "D" if score >= 60
   "F" otherwise

2) Write a program to check if a year is a leap year
   Rules: Leap year if divisible by 4, but not by 100 unless also divisible by 400

3) Write a program that reads a number and prints:
   "Positive and Even" if number > 0 and even
   "Positive and Odd" if number > 0 and odd
   "Negative" if number < 0
   "Zero" if number == 0

4) Write a program that takes 3 sides of a triangle and determines:
   "Equilateral" if all sides equal
   "Isosceles" if exactly two sides equal
   "Scalene" if all sides different
   "Not a triangle" if any side >= sum of other two sides

5) Write a program that takes two numbers and an operator (+, -, *, /)
   Perform the operation and display result
   Handle division by zero with appropriate message

6) Write a program that categorizes age:
   "Child" if age < 13
   "Teen" if 13 <= age < 20
   "Adult" if 20 <= age < 60
   "Senior" if age >= 60

7) Write a program that checks if a character is a vowel or consonant
   Consider both uppercase and lowercase letters

8) Write a program that checks password strength:
   "Weak" if length < 6
   "Medium" if length >= 6 and contains only letters or only digits
   "Strong" if length >= 8 and contains both letters and digits

9) Write a program to find roots of quadratic equation $ax^2 + bx + c = 0$
   Consider all cases: real distinct, real equal, and complex roots

10) Calculate electricity bill based on units consumed:
    First 100 units: ₹5/unit
    Next 100 units: ₹7/unit
    Above 200 units: ₹10/unit
    Add 18% GST if bill > 1000

11) Write a program using for loop to calculate sum of first N natural numbers
12) Write a program using while loop to print multiplication table of a given number

13) Write a program to print first N Fibonacci numbers
14) Write a program to check if a number is prime using a loop
15) Write a program to calculate factorial of a number using for loop
16) Write a program to print the following pattern:
    1
    2 2
    3 3 3
    4 4 4 4
    5 5 5 5 5
17) Write a program to reverse a given number using while loop
    Example: Input: 1234, Output: 4321
18) Write a program to check if a string is palindrome using loop
19) Write a program to check if a number is Armstrong number
    Example: $153 = 1^3 + 5^3 + 3^3 = 153$
20) Write a program to find GCD of two numbers using Euclidean algorithm
21) Write a function that takes a list and returns:
    a. Sum of all elements
    b. Maximum element
    c. Minimum element
    d. Average
22) Write a function that takes a string and returns a dictionary
    with words as keys and their frequencies as values
23) Write a function that takes student records as tuples (name, marks1, marks2, marks3)
    and returns a list of students with average marks above a given threshold
24) Write functions for:
    a. Union of two sets
    b. Intersection of two sets
    c. Difference of two sets
    d. Check if one set is subset of another
25) Implement stack operations using list:
    a. push(item)
    b. pop()
    c. peek()
    d. is_empty()
    e. size()
26) Write functions for:
    a. Matrix addition
    b. Matrix multiplication
    c. Matrix transpose
27) Write a recursive function to find nth Fibonacci number

28) Using list comprehension, create functions that:

a. Generate squares of first N natural numbers

b. Extract even numbers from a list

c. Convert list of Celsius to Fahrenheit

29) Write a function create_email() with parameters:

a) first_name, last_name,
b) domain="gmail.com"
Return email in format: first.last@domain
Handle case sensitivity

30) Write lambda functions for:

i. Square a number
ii. Check if number is even
iii. Concatenate two strings with a space
Use these with map(), filter(), and reduce()